

Freight Rate Prediction Challenge

Machine Learning Engineer Assessment

Modeling, Validation, Data Quality, and Prediction Report

Final model: CatBoost Regressor

Development data: 48,000 labeled loads

Required validation predictions: 12,000 loads

December scenario: 31 daily predictions

1. Executive Summary

This assessment develops a freight-rate regression model using the supplied labeled development data and produces predictions for the 12,000 unlabeled loads in validation.csv. The development data contains 48,000 labeled loads from January through October 2025, while the required validation period covers November and December 2025.

A chronological validation strategy was selected instead of a random split because the final prediction task is explicitly out-of-time. January–August 2025 was used for model fitting and September–October 2025 was reserved as an internal validation period. This design better reflects the temporal structure of the assessment and reduces the risk of optimistic validation caused by mixing future observations into training.

Exploratory analysis showed that distance is the dominant observable driver of posted freight rate. The Pearson correlation between distance and posted_rate is approximately 0.909. Additional signal comes from geographic features, equipment type, weight, market index, and quote signal. The final model uses CatBoost because it captures nonlinear relationships and interactions while handling categorical variables such as pickup, delivery, and equipment directly.

On the internal September–October validation set, CatBoost achieved an MAE of \$122.40 and RMSE of \$636.02, outperforming the evaluated XGBoost baseline (MAE \$208.78, RMSE \$692.66). After model selection, the final CatBoost model was retrained using all 48,000 labeled development rows and used to generate the required 12,000 validation predictions.

2. Dataset Overview

File	Rows	Columns	Role
train_test.csv	48,000	14	Labeled development data
validation.csv	12,000	13	Final prediction data
validation_predictions_template.csv	12,000	2	Required prediction format
december_chart_inputs.csv	31	7	Fixed December scenario

The development target is posted_rate. The labeled development period runs from January 1 through October 31, 2025. The validation period runs from November 1 through December 31, 2025.

The target posted_rate has a mean of approximately \$2,373.98, median of \$2,030.76, standard deviation of \$1,486.49, and range from \$57.22 to \$25,533.00.

Distance has a strong positive relationship with posted_rate (Pearson $r = 0.9085$). The mean rate per mile (posted_rate / distance) is approximately \$2.22/mile, with a median of \$2.15/mile.

3. Exploratory Data Analysis

The exploratory analysis focused on target distribution, route characteristics, equipment mix, temporal coverage, missingness, anomalous values, and relationships between numerical variables and posted_rate.

- Distance is the strongest simple numerical predictor of posted_rate, with correlation $r = 0.9085$.
- The target is right-skewed and contains high-rate observations, with the maximum posted rate reaching \$25,533.
- Dry Van represents approximately 56.7% of loads, Reefer 25.1%, and Flatbed 18.2%.

- Mean RPM is approximately \$2.12 for Dry Van, \$2.38 for Reefer, and \$2.29 for Flatbed.
- The operational road distance and calculated straight-line geographic distance are highly correlated ($r \approx 0.9995$), but both were retained because they represent different concepts.
- The data contains temporal coverage suitable for an out-of-time validation design.

4. Data Quality and Cleaning

Several quality issues were identified before model training. Cleaning and imputation were performed in a leakage-aware manner: statistics used for the internal validation stage were fitted only on the January–August training portion and then applied to September–October.

Issue	Observed	Treatment
Negative weights	292 train rows; 145 validation rows	Converted weight to absolute value
Missing weight	300 train rows; 165 validation rows	Equipment-level median, then overall training median
Missing market_index	374 train rows; 249 validation rows	Date/equipment-based training medians with overall fallback
Unseen validation cities	8 additional cities	Use spatial/geographic features rather than relying only on city categories

The negative weights were treated as data-quality anomalies rather than meaningful negative physical weights. Absolute values were used before modeling. Missing-value statistics were derived from the relevant training data rather than from the validation target period.

5. Feature Engineering

The feature pipeline included the following groups:

- Temporal features: year, month, day of month, day of week, day of year, and week of year.
- Route geometry: latitude/longitude differences, absolute coordinate differences, midpoint latitude/longitude, and haversine geographic distance.
- Operational variables: road distance, equipment, and weight.
- Derived freight features: weight per mile.
- Market interactions: distance $\times$ market_index, distance $\times$ quote_signal, and market_index $\times$ quote_signal.

Feature importance from the CatBoost validation model showed that geographic and distance-related variables dominate the prediction task. The five highest-importance features were geo_distance (32.07%), distance (19.20%), distance_quote (12.55%), distance_market (10.58%), and abs_lon_diff (7.95%). Together they account for roughly 82% of the model's feature importance.

6. Training and Validation Strategy

A chronological split was used to simulate the assessment's real prediction setting. The split was:

Partition	Period	Rows	Purpose
Training	Jan 1 – Aug 31, 2025	38,477	Fit models and preprocessing
Internal validation	Sep 1 – Oct 31, 2025	9,523	Model selection and error analysis
Final assessment validation	Nov 1 – Dec 31, 2025	12,000	Required predictions; labels withheld

Preprocessing statistics for the internal validation experiment were fitted using only the January–August training partition. The September–October observations were transformed using those training-derived statistics. This prevents information from the validation period from influencing the preprocessing stage.

After selecting the final configuration, the CatBoost model was retrained on all 48,000 labeled development rows. The resulting model was then applied to the 12,000 assessment validation rows.

7. Model Development and Comparison

Two gradient-boosting approaches were evaluated during development: CatBoost and XGBoost. Both are suitable for nonlinear tabular regression and can model interactions between route, equipment, and operational variables.

Model	Internal MAE	Internal RMSE	Decision
CatBoost	\$122.40	\$636.02	Selected
XGBoost	\$208.78	\$692.66	Baseline / not selected

CatBoost reduced internal validation MAE by approximately 41% relative to the XGBoost baseline. This provided the main quantitative basis for selecting CatBoost for the final submission.

7.1 Final CatBoost Configuration

Parameter	Value
iterations	2000
depth	7
learning_rate	0.05
loss_function	RMSE
eval_metric	MAE
l2_leaf_reg	3
random_strength	1
bagging_temperature	0
random_seed	42

The final model was trained on the complete 48,000-row labeled development dataset after the model-selection stage. Its training-set MAE of approximately \$97.04 is not reported as validation performance because it is a training metric.

8. Internal Validation Error Analysis

The CatBoost internal validation MAE was \$122.40, while the median absolute error was approximately \$46.30. The 90th and 95th percentiles of absolute error were approximately \$158.86 and \$232.40 respectively.

Distance group	MAE
< 500 miles	\$59.47
500–1,000 miles	\$81.10
1,000–1,500 miles	\$126.81
1,500–2,000 miles	\$195.48
2,000–2,500 miles	\$216.59
2,500+ miles	\$246.55

The error analysis indicates that long-haul loads are substantially harder to predict and contribute strongly to RMSE. The largest errors were generally large underpredictions on unusually high-rate, long-distance loads.

9. Final Validation Predictions

The final CatBoost model was applied to every row in validation.csv. The resulting file contains all 12,000 required load_id values and a predicted_rate value for each row.

- Prediction count: 12,000
- Missing/invalid predictions: none
- Prediction range: approximately \$12.85 to \$9,559.99
- Mean predicted rate: approximately \$2,438.50

The completed submission file follows the required two-column schema: load_id,predicted_rate.

10. December Scenario Predictions

The December scenario contains 31 daily rows for a fixed Lexington → Fort Wayne route. The input file does not contain pickup/delivery coordinates, market_index, or quote_signal, although these variables are part of the main model feature set.

For this scenario, coordinates for Lexington and Fort Wayne were mapped from the main dataset. Missing market and quote information was supplied using October historical medians by equipment, with an overall fallback. This keeps the scenario inputs consistent with the final model's expected feature structure.

The resulting December predictions range from approximately \$734.80 to \$796.19, with a mean of approximately \$758.25.

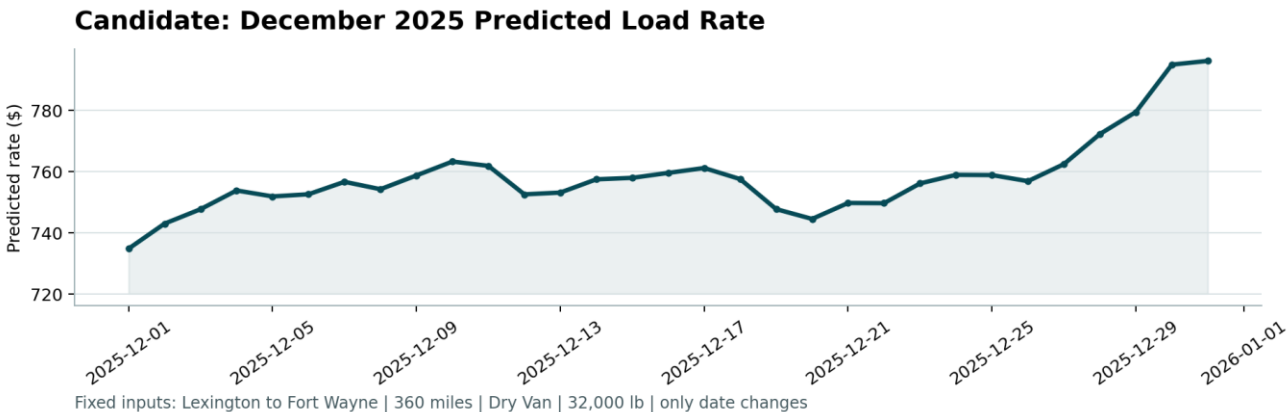


Figure 1. Official December 2025 predicted load-rate chart generated by the provided score.py scorer.

10.1 Fixed December Prediction Chart

11. Official Scorer Verification

The provided score.py was executed against the completed 12,000-row validation prediction file and the completed 31-row December scenario file.

Check	Result
Validation predictions	Validated 12,000 final predictions
December predictions	Validated 31 fixed December predictions
December chart	Created scorer_results/candidate_december.png
Official hidden-set metrics	Not available from scorer; calculated by Spotter after submission

12. Limitations and Considerations

- The official November–December target values are withheld, so the final assessment-set MAE/RMSE cannot be calculated locally.
- The December scenario lacks several model inputs, requiring documented proxy/imputation values.
- Extreme high-rate long-haul loads remain the largest source of internal validation error.
- The final model is optimized for the supplied assessment data and should be monitored before production deployment on materially different freight markets.
- The validation methodology intentionally prioritizes temporal realism over random cross-validation because the assessment's prediction period follows the labeled development period.

13. Conclusion

The completed solution uses a leakage-aware chronological validation strategy, explicit data-quality handling, route/geographic feature engineering, and CatBoost regression for freight-rate prediction. CatBoost provided substantially stronger internal validation performance than the evaluated XGBoost baseline. The final model was then retrained on all available labeled development data and used to generate the required 12,000 validation predictions and 31 December scenario predictions.

The official scorer confirmed that both prediction files satisfy the required row counts and formats. The final hidden validation metrics remain the responsibility of the assessment platform and are not inferred from the local scorer.

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14. Reproducibility and Submission Artifacts

- GitHub repository: source code, notebook/scripts, requirements, and run instructions.
- validation_predictions.csv: final 12,000-row submission.
- December prediction file: 31-row scenario predictions.
- Final CatBoost model and preprocessing artifacts saved separately for reproducibility.
- Official scorer output: candidate_december.png.

All preprocessing and model-selection decisions should be reproduced from the training data before generating final assessment predictions.

End of Report